

Bauchi State University, Gadau

Faculty of Science – Department of Chemistry

Course Codes: CHM101 – General Chemistry & CHM107 – Practical Chemistry

Academic Session: 2023/2024 – First Semester Examination

Time Allowed: 1 Hour (CHM101) / 2 Hours (CHM107)

Instruction: ANSWER ALL QUESTIONS

Questions 1–60

1. When two or more elements combine chemically, they form

- (A) Lipid
- (B) Atom
- (C) Elements
- (D) Compound ☒

2. Which of the following has a positive charge?

- (A) Electron
- (B) Proton ☒
- (C) Nucleus
- (D) Atom

3. Based on its composition, matter can be classified into three types with the exception of

- (A) Element
- (B) Compound
- (C) Mixture
- (D) Molecule ☒

4. Ethyne (C_2H_2) is not a type of hybridization

- (A) sp^3 ☒
- (B) sp^2
- (C) sp
- (D) None

5. The shape of ethene (CH_2) is treated as

- (A) sp^3 hybridized to be tetrahedral shape and angle ☒
- (B) Lone pairs of electrons are responsible for the shape
- (C) The shape and size of s and p are different
- (D) The carbon and hydrogen are of different sizes

6. Which of these statements is not true

- (A) Valence forces are either attractive or repulsive
- (B) Electrostatic forces are between charged particles
- (C) Covalent forces are between atoms sharing electrons through bonding

(D) Ionic bonding is a type of covalent bonding ✓

7. How many unhybridized P orbitals are in ethyne?

(A) 1

(B) 2 ✓

(C) 3

(D) 4

8. The particles that are closely packed together are

(A) Liquid particles

(B) Solid particles ✓

(C) Gaseous particles

(D) Dust particles

9. Sodium chloride dissolves in water because it is

(A) Non-polar bond

(B) Polar bond ✓

(C) Hydrogen bond

(D) None of the above

10. The type of bonding that involves sharing of outermost electrons is

(A) Electrovalent bonding

(B) Covalent bonding ✓

(C) Coordinate covalent bonding

(D) Metallic bonding

11. states that the volume of a given mass of gas is inversely proportional to the pressure provided that the temperature remains constant.

(A) Charles law

(B) Boyle's law ✓

(C) Graham's law

(D) Dalton's law

12. The chemical behavior of gases was studied by

(A) Robert Boyle ✓

(B) Joseph Louis Gay-Lussac

(C) Sir Humphry Davy

(D) Antoine Lavoisier

13. When old bonds are broken and new bonds are formed to create a new substance, the process is

(A) New process

(B) New compound

(C) Chemical reaction ✓

(D) Chemical equation

14. Which of the following equations is balanced?

- (A) $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- (B) $\text{C}_{14}\text{H}_{10}\text{O}_2 + \text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$
- (C) $\text{CH}_4 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$ ✓
- (D) None

15. A mathematical statement showing product formation from reactants under certain conditions is called

- (A) New process
- (B) New compound
- (C) Chemical reaction
- (D) Chemical equation ✓

16. A balanced equation showing quantitative relationships of reactants and products is called

- (A) Stoichiometry ✓
- (B) Balanced Equation
- (C) Combustion reaction
- (D) None

17. The function of Manganese (IV) oxide in a reaction is

- (A) Catalyst ✓
- (B) Reactant
- (C) Product
- (D) Solvent

18. Which of the following is not a personal protective equipment?

- (A) Burette ✓
- (B) Eye Goggle
- (C) Lab Coat
- (D) Nose Mask

19. One of these is used in volumetric analysis

- (A) Pipette ✓
- (B) Spatula
- (C) Conical Flask
- (D) Glass rod

20. Which of the following is used in acid spillage?

- (A) Sodium carbonate ✓
- (B) Sodium chloride
- (C) Potassium chloride
- (D) None of the above

21. Which of the following chemicals is corrosive?

- (A) NaOH ✓

- (B) NaCl
- (C) K_2CO_3
- (D) Na_2CO_3

22. Liquid chemical waste is disposed in

- (A) Dust bin
- (B) Water distiller
- (C) The hood
- (D) Sink ☒

23. Toxic fumes are expelled using

- (A) Fume hood ☒
- (B) Sink
- (C) Oven
- (D) None of the above

24. One of these is not used when heating

- (A) Wire Gauze
- (B) Bunsen Burner
- (C) Clay Triangle
- (D) Burette ☒

25. To determine the density of an aqueous NaCl solution, which measurements are required?

- (A) Mass of the solution
- (B) Volume of the solution
- (C) Both mass and volume ☒
- (D) Temperature

26. If 100 mL of NaCl solution has a mass of 105.0 g, what is the density?

- (A) 0.942 g/mL
- (B) 1.054 g/mL ☒
- (C) 1.5 g/mL
- (D) 0.5 g/mL

27. Density of pure water at 25°C is 0.997 g/mL. NaCl solution has density 1.025 g/mL. Conclusion?

- (A) Less dense than water
- (B) More dense than water ☒
- (C) Same density
- (D) Cannot determine

28. Density of 1.5M NaCl solution at 20°C is 1.072 g/mL. Mass of 500 mL?

- (A) 536 g ☒
- (B) 214.4 g
- (C) 2.144 g

(D) 0.0214 g

29. An aqueous NaCl solution has density 1.1 g/mL. If mass is 4.2 g, volume is:

(A) 400 mL

(B) 4000 mL

(C) 3.8 mL ☒

(D) 0.4 mL

30. NaOH solution (1.02 g/mL) mixed with 200 mL water. Final density?

(A) 1.067 g/mL ☒

(B) 0.8 g/mL

(C) 1.133 g/mL

(D) 0.8 g/M

31. Primary purpose of standardizing NaOH with oxalic acid?

(A) Determine exact concentration ☒

(B) Neutralize NaOH

(C) Prepare buffer

(D) Convert NaOH to sodium salt

32. Balanced equation for NaOH and oxalic acid?

(A) $\text{H}_2\text{C}_2\text{O}_4 + \text{NaOH} \rightarrow \text{NaC}_2\text{O}_4 + \text{H}_2\text{O}$

(B) $\text{H}_2\text{C}_2\text{O}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{C}_2\text{O}_4 + 2\text{H}_2\text{O}$ ☒

(C) $\text{NaOH} + \text{H}_2\text{C}_2\text{O}_4 \rightarrow \text{NaHC}_2\text{O}_4 + \text{H}_2\text{O}$

(D) $2\text{NaOH} + \text{H}_2\text{C}_2\text{O}_4 \rightarrow \text{Na}_2\text{C}_2\text{O}_4 + \text{NaHC}_2\text{O}_4 + \text{H}_2\text{O}$

33. Role of oxalic acid in NaOH standardization?

(A) Primary standard ☒

(B) Buffer

(C) Indicator

(D) Solvent

34. Best indicator for NaOH titration with oxalic acid?

(A) Phenolphthalein ☒

(B) Methyl orange

(C) Bromothymol green

(D) Methyl red

35. To calculate NaOH concentration after standardization, you need:

(A) Volume of NaOH and mass of oxalic acid

(B) Volume of oxalic acid and its concentration

(C) Volume of NaOH and concentration of oxalic acid ☒

(D) Mass and concentration of oxalic acid

36. Molar mass of oxalic acid = 90.04 g/mol. Mass for 250 mL of 0.1M solution?

- (A) 2.25 g ☒
- (B) 4.50 g
- (C) 9.00 g
- (D) 1.80 g

37. Reaction between NaOH and oxalic acid is:

- (A) Exothermic
- (B) Endothermic
- (C) Neutralization
- (D) Both A and C ☒

38. Factors for selecting a standard:

- (A) High purity
- (B) Stability
- (C) Ease of handling
- (D) All of the above ☒

39. In redox reaction: $5\text{Fe}^{2+} + \text{MnO}_4^- + 8\text{H}^+ \rightarrow 5\text{Fe}^{3+} + \text{Mn}^{2+} + 4\text{H}_2\text{O}$, Mn changes from:

- (A) +7 to +2 ☒
- (B) +2 to +7
- (C) +2 to +3
- (D) +3 to +2

40. If 25 mL of 0.02 M KMnO_4 solution requires 23.5 mL of 0.05 M $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2$ solution for complete reaction, what is the actual molarity of KMnO_4 ?

- (A) 0.0188 M
- (B) 0.0196 M ☒
- (C) 0.0165 M
- (D) 0.0168 M

41. Approximately what mass of KMnO_4 is required to make 500 mL of 0.02 M solution?

- (A) 1.58 g
- (B) 3.16 g ☒
- (C) 4.74 g
- (D) 6.32 g

42. What is the purpose of swirling constantly during titration?

- (A) To ensure thorough mixing ☒
- (B) To remove bubbles
- (C) To equilibrate temperature
- (D) To prevent loss due to high concentration

43. A 0.1 M oxalic acid solution is prepared by dissolving 3.603 g of oxalic acid dihydrate ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in water and diluting to 500 mL. If 25.0 mL of this solution is required to titrate 25.0 mL of NaOH solution, what is the concentration of the NaOH solution?

☒ 0.1 M (based on 1:1 molar ratio)

44. If 30.0 mL of NaOH solution is titrated with 24.0 mL of 0.09 M oxalic acid, what is the concentration of NaOH?

(A) 0.0720 M

(B) 0.0900 M

(C) 0.1080 M ☒

(D) 0.1350 M

45. Express 25 cm^3 in dm^3

(A) 0.025 dm^3 ☒

(B) 0.25 dm^3

(C) 250 dm^3

(D) 2.5 dm^3

46. Express 0.2 g/dm^3 NaOH in mol/dm^3 (Molar mass = 40 g/mol)

(A) 0.0055 M ☒

(B) 0.01 M

(C) 4.0 M

(D) 0.04 M

47. A student obtained titre values: 25.10 mL, 24.95 mL, 26.30 mL, and 25.00 mL. What is the exact titre value?

(A) 25.00 ☒

(B) 25.13

(C) 25.30

(D) None of the above

48. Which of the following is used to measure accurate volume in titration?

(A) Beaker

(B) Measuring cylinder

(C) Burette ☒

(D) Test tube

49. Which of the following is a primary standard substance?

(A) Sodium hydroxide

(B) Oxalic acid ☒

(C) Hydrochloric acid

(D) Sulfuric acid

50. Which of the following is NOT a property of a good primary standard?

(A) High purity

(B) Low reactivity

(C) Hygroscopic nature ☒

(D) Stable in air

51. What is the color change of phenolphthalein in titration with NaOH and oxalic acid?

(A) Pink to colorless ☒

(B) Colorless to pink

(C) Yellow to red

(D) Red to yellow

52. Which glassware is used to prepare standard solutions accurately?

(A) Beaker

(B) Volumetric flask ☒

(C) Conical flask

(D) Test tube

53. Which of the following is used to transfer solid chemicals?

(A) Pipette

(B) Spatula ☒

(C) Burette

(D) Funnel

54. Which of the following is used to rinse burette before titration?

(A) Water

(B) Acid

(C) Base solution ☒

(D) Alcohol

55. Which of the following is used to detect end point in acid-base titration?

(A) Thermometer

(B) Indicator ☒

(C) pH meter

(D) Conductivity meter

56. Which of the following is used to hold burette during titration?

(A) Retort stand ☒

(B) Tripod stand

(C) Clamp

(D) Wire gauze

57. Which of the following is used to heat substances in the lab?

(A) Bunsen burner ☒

(B) Fume hood

(C) Oven

(D) Refrigerator

58. Which of the following is used to support crucible during heating?

- (A) Wire gauze
- (B) Clay triangle ☒
- (C) Tripod stand
- (D) Beaker

59. Which of the following is used to measure temperature?

- (A) Thermometer ☒
- (B) Barometer
- (C) Hygrometer
- (D) Manometer

60. Which of the following is used to dry substances in the lab?

- (A) Oven ☒
- (B) Refrigerator
- (C) Fume hood
- (D) Desiccator